(8) A tow truck drags a stalled car along a road. The chain makes an angle of 30° with the road, and the tension in the chain is 1500 N. Use the techniques developed in this chapter to determine how much work is done by the truck in pulling the car 1 kilometer.

(9) Compute the cross product $(3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 3\hat{\mathbf{k}}) \times (3\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$.

(10) Find the area of the triangle with vertices $P(1, 0, 2)$, $Q(3, 3, 3)$, and $R(7, 5, 8)$.

(11) Find the volume of the pyramid with vertices $P(3, 0, 1)$, $Q(-1, 2, 5)$, $R(5, 1, -1)$, and $S(0, 4, 2)$.